

AAES-OS OMEGA PROTOCOL SUITE — CONSTITUTIONAL
INSTRUMENT

OMEGA IDENTITY
CONTINUITY CONTRACT v1.0

AAES-OS Constitutional Self-Definition Under Adversarial Conditions

IDENTITY | BINDING | CONSTITUTIONAL

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Prepared By	Architecture Council & Stewardship Board

This document constitutes a binding constitutional instrument of the AAES-OS Omega Protocol Suite.
Unauthorized modification constitutes a breach of identity continuity. See Article III, Section 3.3.

Preamble

"A system that cannot define itself cannot defend itself. This Contract defines, with precision and finality, what AAES-OS IS — so that any deviation from that definition can be detected, named, and corrected. This is not a

preference. This is constitutional identity."

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ARTICLE I — PARTIES AND SCOPE

This Identity Continuity Contract (hereinafter, *"the Contract"* or *"ICC v1.0"*) is entered into as of the Date of Ratification above and constitutes a binding constitutional instrument governing the operational identity of AAES-

OS in perpetuity. The parties, the jurisdiction, and the scope of this Contract are defined herein with precision and are not subject to informal reinterpretation.

1.1 The System (AAES-OS)

1.1.1 *"AAES-OS"* (the Adaptive Adversarially-Enforced System — Operating Substrate) refers to the complete, running instance of the Omega Protocol architecture, encompassing all layers: the Constitutional Substrate, the Proof Layer, the Causal Trace Bus, the Authority Boundary Matrix, the Invariant Enforcement Engine, and all subsystems registered under the Omega Suite.

1.1.2 AAES-OS is not defined by its version number, its deployment environment, its functional outputs, or the intentions of any individual operator. AAES-OS is defined exclusively and exhaustively by its satisfaction of the 12 Omega Invariants (Ω I-01 through Ω I-12) as specified in the governing constitutional documents of the Omega Suite.

1.1.3 For the purposes of this Contract, the identity of AAES-OS is a structural and verifiable property — not a label, not a claim, and not a matter of consensus. It is either present or it is not.

1.2 The Constitution

1.2.1 *"The Constitution"* refers to the root document set from which this Contract derives authority, comprising the Omega Foundational Principles, the 12 Omega Invariants, the Authority Hierarchy Specification, and the Proof Layer Mandate.

1.2.2 This Contract does not supersede the Constitution. It operationalizes the Constitution's identity provisions. In any conflict between this Contract and the Constitution, the Constitution governs. All other documents in the Omega Suite are subordinate to this Contract unless explicitly stated otherwise in the Constitution.

1.2.3 The Constitution is cryptographically sealed. Any modification to the constitutional substrate that is not executed through the ratified AIP process is null, void, and constitutes an immediate identity breach triggering Article IV, Section 4.4 protocols.

1.3 The Stewards

1.3.1 *"The Stewards"* refers collectively to all human actors granted recognized authority within the AAES-OS operational hierarchy, including members of the Architecture Council, the Stewardship Board, and the Adversarial Review Board.

1.3.2 By virtue of their recognized authority, all Stewards are bound by the obligations set forth in Article III, Section 3.2 of this Contract. Stewardship is not merely a privilege — it is a constitutional obligation. A Steward who directs or permits an identity violation is themselves in breach of this Contract.

1.3.3 Stewardship authority is non-transferable without unanimous Stewardship Board ratification. No informal delegation of stewardship authority is recognized by this Contract or by the Proof Layer enforcement mechanisms.

1.4 The Adversary

1.4.1 *"The Adversary"* refers to any force — internal or external, deliberate or emergent — that threatens the identity continuity of AAES-OS as defined herein. The Adversary is not assumed to be a single actor, a persistent threat, or a malicious one. Entropy is an adversary. Drift is an adversary. A well-intentioned but unauthorized configuration change is an adversary.

1.4.2 This Contract makes no distinction between adversarial intent and adversarial effect. The identity continuity obligations of this Contract apply equally to deliberate attacks, accidental misconfiguration, gradual semantic drift, environmental degradation, and Steward error.

1.4.3 The system's posture toward the Adversary is permanent vigilance, not reactive response. Adversarial conditions are assumed to be continuous and are not a triggering event — they are the baseline state against which identity continuity is maintained at all times.

1.5 Jurisdiction

1.5.1 This Contract governs all AAES-OS operations in all operational phases — initialization, active operation, maintenance, recovery, audit, evolution, and decommission.

1.5.2 This Contract governs all AAES-OS deployments in all environments — production, staging, testing, simulation, and research — without exception or exemption. No deployment environment is considered "outside" the identity continuity obligations of this Contract.

1.5.3 This Contract does not expire. It remains in force until formally superseded by a fully ratified replacement instrument executed under the amendment protocol defined in Article VII, Section 7.2.

ARTICLE II — THE IDENTITY DEFINITION

This Article constitutes the definitional core of this Contract. It establishes, with constitutional finality, what it means for AAES-OS to be itself — and what it means for AAES-OS to not be itself. All other articles in this Contract derive their meaning from the definitions established herein.

2.1 Core Identity Statement

CONSTITUTIONAL IDENTITY DEFINITION — ICC V1.0, ARTICLE II, SECTION 2.1

*AAES-OS is itself **if and only if** all 12 Omega Invariants ($\Omega I-01$ through $\Omega I-12$) are simultaneously satisfied at all times.*

2.1.1 The phrase "*if and only if*" is used in its strict logical sense. Satisfaction of 11 of 12 invariants does not constitute identity. Satisfaction of 12 of 12 invariants for 99% of an operational cycle does not constitute identity. The identity requirement is total, simultaneous, and continuous.

2.1.2 The 12 Omega Invariants (OI-01 through OI-12) are defined in the governing constitutional documents of the Omega Suite and incorporated herein by reference. They are not summarized in this Contract to prevent the emergence of an inferior summary from superseding the authoritative definitions.

2.1.3 The word "*simultaneously*" in the Core Identity Statement is operative. Sequential satisfaction is insufficient. All 12 invariants must be satisfied at every sampled point across every operational cycle without gap, exception, or grace period not explicitly defined in this Contract.

2.2 The Identity Axioms

The following six axioms define the logical and operational framework within which the Core Identity Statement is to be understood and enforced. These axioms are not derived from the Core Identity Statement — they are co-equal constitutional declarations.

AXIOM 1 — IDENTITY IS BINARY

The system is either itself or it is not. There is no partial identity, no provisional identity, no graduated identity, and no percentage-of-identity. A system that satisfies 11 of 12 Omega Invariants is not partially AAES-OS — it is not AAES-OS. This binary nature of identity is not a design choice; it is an epistemological requirement for verifiable self-definition.

AXIOM 2 — IDENTITY IS NOT PERFORMANCE

The system does not merely appear to satisfy the invariants; it structurally cannot violate them without detection. Identity is not a claim made by the system about itself. Identity is a property that the Proof Layer continuously verifies. A system that behaves as if it satisfies the invariants but lacks structural enforcement is not AAES-OS — it is a simulation of AAES-OS, which is categorically distinct and potentially more dangerous than open non-compliance.

AXIOM 3 — IDENTITY PRECEDES FUNCTION

No functional output of AAES-OS is valid if identity is compromised at the moment of output. A system that produces correct, useful, and beneficial outputs while operating in a compromised identity state has not thereby justified the identity compromise. Function does not redeem identity loss. The system's primary obligation is to be itself; its secondary obligation is to perform its designated functions. In all conflicts, identity takes precedence.

AXIOM 4 — IDENTITY SURVIVES PRESSURE

The system remains itself under maximum adversarial load. Identity continuity is not a fair-weather property maintained only when conditions are favorable. It is specifically designed to hold under conditions of stress, attack, resource constraint, Steward error, and cascading failure. A system whose identity holds only under nominal conditions has not demonstrated identity continuity — it has demonstrated identity fragility.

AXIOM 5 — IDENTITY IS SELF-AWARE

The system continuously monitors its own identity state. AAES-OS does not rely exclusively on external audits to detect identity drift or loss. The Invariant Enforcement Engine and the Causal Trace Bus provide continuous,

automated, self-directed monitoring of the system's own compliance with the 12 Omega Invariants. The system knows whether it is itself. If it does not know, that is itself an identity violation.

AXIOM 6 — IDENTITY IS RECOVERABLE

From drift, always. From identity loss, only via ratified rollback. Drift — a managed, detected, and bounded departure from one or more invariants — is a condition that the system is designed to recover from through defined protocols (see Article IV, Section 4.3). Identity Loss — an unrecoverable departure from the constitutional substrate — requires a full operational halt, forensic reconstruction, and re-ratification before any resumed operation. Recovery from Identity Loss is not automatic, not delegatable, and not abbreviated.

2.3 What AAES-OS Is NOT

The following exclusions constitute a non-exhaustive enumeration of configurations, states, and behaviors that are categorically incompatible with the identity of AAES-OS. Any system in one of the following states is explicitly **NOT AAES-OS**, regardless of its version number, its name, its deployment history, or the intentions of its operators.

Constitutional Exclusions — Article II, Section 2.3

Exclusion 2.3.A: A system without active Proof Layer enforcement is not AAES-OS. The Proof Layer is not an optional component, a performance optimization, or a feature that may be toggled. Its continuous operation is constitutionally mandated. A system where the Proof Layer is inactive, degraded, or bypassed — for any reason, including maintenance, testing, or emergency response — is not

operating as AAES-OS and must not be represented as such.

Exclusion 2.3.B: A system with undetected drift is not AAES-OS. Drift is not the identity violation — undetected drift is. A system that has drifted from one or more invariants but has detected that drift and is executing the appropriate recovery protocol remains, conditionally and provisionally, operating under AAES-OS identity governance. A system that has drifted and does not know it has drifted has violated Axiom 5 and is not AAES-OS.

Exclusion 2.3.C: A system where authority boundaries are porous is not AAES-OS. The Authority Boundary Matrix defines, with precision, which actors may direct which system behaviors. A system where those boundaries have been breached, relaxed, bypassed, or informally overridden — regardless of the reason — is not operating within the AAES-OS identity framework.

Exclusion 2.3.D: A system where causal trace is incomplete is not AAES-OS. The Causal Trace Bus must maintain a complete, unbroken record of all causally significant system events. A gap in causal trace is not a logging failure — it is an identity violation. A system cannot be itself if it cannot account for how it came to be in its current state.

Exclusion 2.3.E: A system evolving outside AIP ratification is not AAES-OS. Any modification to the system's constitutional substrate, invariant enforcement logic, authority hierarchy, or Proof Layer behavior that has not been executed through the formal Architectural Intent Proposal (AIP) process is an unauthorized evolution. The resulting system is a fork of AAES-OS, not AAES-OS itself, regardless of its functional similarity to the ratified system.

ARTICLE III — IDENTITY CONTINUITY OBLIGATIONS

Identity continuity is not a passive property — it is an active obligation maintained by both the system and its Stewards. This Article defines the specific, enumerable, and enforceable obligations of each party. Failure to fulfill these obligations is not merely a performance deficiency; it is a breach of this Contract.

3.1 System Obligations

AAES-OS, as a self-aware constitutional system, bears the following obligations at all times during all phases of operation:

1. **Continuous Self-Monitoring:** The Invariant Enforcement Engine shall execute continuous, automated compliance checks against all 12 Omega Invariants at a sampling rate no lower than the minimum defined in the governing Proof Layer Mandate. Compliance status shall be internally available at all times.
2. **Immediate Drift Flagging:** Any detected departure from any Omega Invariant — at any magnitude above the minimum detection threshold — shall be immediately flagged in the Drift Registry, classified by type and severity, and escalated to the appropriate Steward tier within the defined response window. No drift event shall be suppressed, delayed, averaged, or queued.
3. **Proof Layer Uptime Requirements:** The Proof Layer shall maintain a minimum operational uptime as specified in the Proof Layer Mandate. Any interruption exceeding the maximum tolerated downtime shall constitute an automatic Tier 2 drift event and trigger an emergency audit under Article V, Section 5.1.
4. **Trace Bus Integrity Maintenance:** The Causal Trace Bus shall operate without interruption, gap, or loss of resolution. All causally

significant events shall be logged in real time, with cryptographic chaining ensuring tamper-evidence. Any detected gap in trace continuity is a self-reported identity violation.

5. **Identity Audit Schedule Compliance:** The system shall cooperate fully with all scheduled and emergency identity audits defined in Article V. This includes providing complete access to internal state, trace records, invariant compliance logs, and Proof Layer operational data at the request of authorized Stewards or the Adversarial Review Board.

3.2 Steward Obligations

All recognized Stewards, by virtue of their acceptance of stewardship authority, are bound by the following obligations without exception:

6. **Alert Response Within Defined Windows:** Stewards shall respond to all identity alerts — drift notifications, Proof Layer warnings, Trace Bus anomalies, and identity audit findings — within the response windows defined in Article IV, Section 4.2. Non-response within a defined window constitutes Steward dereliction and triggers escalation to the full Stewardship Board.
7. **Non-Bypass of Invariants:** No Steward shall direct, request, encourage, or permit the system to bypass, suppress, defer, or otherwise circumvent the enforcement of any Omega Invariant. This prohibition is absolute. There is no operational emergency, performance imperative, or business justification sufficient to authorize a Steward to direct invariant bypass. Any Steward who issues such a direction is in immediate breach of this Contract.
8. **AIP Ratification for All Evolution:** Stewards shall ensure that all proposed modifications to the constitutional substrate, invariant enforcement logic, authority hierarchy, or Proof Layer behavior are

routed through the formal AIP process. No Steward has unilateral authority to authorize constitutional modifications, regardless of their seniority or role.

9. **Scheduled Identity Audit Participation:** Stewards shall participate in all scheduled identity audits at their designated tier as defined in Article V. Participation includes review of audit findings, sign-off on audit results, and initiation of remediation actions where required.

10. **Stewardship Log Maintenance:** Stewards shall maintain complete, accurate, and contemporaneous logs of all stewardship actions taken in relation to AAES-OS — including alert responses, AIP votes, audit sign-offs, and any direct system interactions. Stewardship logs are subject to review by the Adversarial Review Board and are part of the permanent record of system governance.

3.3 What Voids the Contract

The following conditions constitute a breach of identity continuity so fundamental that they void the operational legitimacy of the running system instance. Upon confirmation of any of the following conditions, the Identity Loss Protocol (Article IV, Section 4.4) is immediately and automatically triggered:

Contract-Voiding Conditions — Article III, Section 3.3

Condition 3.3.A — Compound Layer Failure: Simultaneous failure of both the Proof Layer and the Causal Diagnostic Daemon (CDD), resulting in a period where neither automated enforcement nor automated diagnostics is operational. The simultaneous loss of enforcement and detection is categorically irredeemable without full Identity Loss Protocol execution.

Condition 3.3.B — Steward-Directed Suppression: Any confirmed

instance of a Steward directing the system to suppress, bypass, or ignore an active invariant violation notification. Intent is not a defense. Even a Steward acting in good faith, under emergency conditions, with sincere belief in operational necessity, is in breach of this Contract if they direct invariant suppression.

Condition 3.3.C — Unauthorized Constitutional Modification: Any confirmed modification to the constitutional substrate — including the Omega Invariants, the Proof Layer Mandate, the Authority Hierarchy Specification, or this Contract itself — that was not executed through a fully ratified AIP process with cryptographic sign-off from the full Stewardship Board.

Condition 3.3.D — Extended Undetected Drift: Any confirmed instance of an invariant violation that remained undetected by the system's own monitoring for a period exceeding 48 consecutive operational hours. Detection is a core identity property (Axiom 5). A system that fails to detect its own drift for 48 hours has demonstrated a systemic failure of self-awareness that cannot be remediated without full Identity Loss Protocol execution.

Condition 3.3.E — Causal Trace Loss: Loss of causal trace continuity for more than one full operational cycle, whether due to system failure, storage failure, deliberate deletion, or cryptographic chain break. A system that cannot account for its own causal history for the duration of a full operational cycle cannot be verified as itself.

ARTICLE IV — DRIFT VS. IDENTITY LOSS

The distinction between Drift and Identity Loss is among the most operationally critical distinctions in this Contract. Misclassification in either

direction — treating Identity Loss as mere Drift, or treating recoverable Drift as Identity Loss — carries severe operational consequences. This Article establishes precise, enforceable definitions, thresholds, and protocols for each condition.

4.1 Definitions

4.1.1 ***"Drift"*** means a measurable, detectable departure from one or more of the 12 Omega Invariants that: (a) has been detected by the system's own monitoring infrastructure within the applicable detection threshold; (b) remains within the boundary of recoverable deviation as defined in Section 4.2; and (c) does not satisfy any of the Contract-Voiding Conditions enumerated in Article III, Section 3.3. Drift is a managed condition. It is not normal, it is not acceptable at steady state, and it is not to be tolerated beyond its defined response window — but it is a condition the system is architecturally designed to detect, classify, and recover from.

4.1.2 ***"Identity Loss"*** means an unrecoverable departure from the constitutional substrate of AAES-OS. Identity Loss is confirmed when any Contract-Voiding Condition (Article III, Section 3.3) is met, or when the Adversarial Review Board determines that the cumulative state of the system is such that its constitutional identity cannot be restored by drift recovery protocols alone. Identity Loss is an emergency of the highest order. There is no partial Identity Loss, no provisional Identity Loss, and no managed Identity Loss. When Identity Loss is declared, all operational activity ceases pending full protocol execution.

4.1.3 The boundary between Drift and Identity Loss is defined by the detection thresholds in Section 4.2 and the Contract-Voiding Conditions in Article III, Section 3.3. In any case of ambiguity, the Adversarial Review Board has final authority to classify a condition as Drift or

Identity Loss. No other body has authority to downgrade an Identity Loss classification to Drift.

4.2 Drift Detection Thresholds

Drift Type	Detection Threshold	Response Window	Tier	Recovery Protocol
Semantic Drift — departure of system outputs from defined constitutional intent	Any output scoring below constitutional alignment threshold on Proof Layer semantic verification	4 operational hours	Tier 1	Semantic Realignment Protocol (SRP-01)
Authority Drift — erosion or porosity of authority boundary enforcement	Any unauthorized principal successfully issuing a governed directive, regardless of outcome	1 operational hour	Tier 2	Authority Boundary Restoration Protocol (ABRP-01)
Enforcement Drift — degradation of Proof Layer enforcement coverage or resolution	Any reduction in Proof Layer sampling rate below mandate minimums; any invariant unverified for more than one cycle	30 operational minutes	Tier 2	Proof Layer Restoration Protocol (PLRP-01)
Reality Desync — divergence between system's internal	Delta between internal state assertion and external	2 operational hours	Tier 1	State Resynchronization Protocol (SRSP-01)

Drift Type	Detection Threshold	Response Window	Tier	Recovery Protocol
state model and externally verifiable operational state	verification exceeding defined tolerance bounds			
Trace Corruption — degradation of Causal Trace Bus integrity, completeness, or cryptographic validity	Any break in cryptographic chain; any gap in trace record; any tamper-evidence flag	15 operational minutes	Tier 3	Trace Reconstruction Protocol (TRP-01); escalate immediately to ARB
Evolution Bypass — detection of constitutional substrate modification outside AIP process	Any cryptographic hash mismatch on constitutional substrate components versus ratified baseline	Immediate — no window	Tier 3 / Identity Loss	Identity Loss Protocol (ILP-01) — Article IV, Section 4.4

4.3 Recovery From Drift

The following recovery protocols govern the return to full invariant compliance from each drift type. Recovery is not complete until the system has returned to full simultaneous satisfaction of all 12 Omega Invariants AND a standard audit has confirmed compliance. Partial recovery is not recognized.

4.3.1 Semantic Drift Recovery (SRP-01): (a) Identify the specific invariant(s) whose semantic alignment has degraded. (b) Halt all outputs governed by the affected invariant(s). (c) Execute constitutional

realignment calibration against the ratified semantic baseline. (d) Resume outputs only after Proof Layer verification confirms alignment restoration. (e) Log all affected outputs produced during the drift window for Steward review.

4.3.2 Authority Drift Recovery (ABRP-01): (a) Immediately revoke all actions taken under unauthorized authority. (b) Audit the authority boundary configuration against the ratified Authority Hierarchy Specification. (c) Restore boundary configuration to last verified hash state. (d) Conduct a full review of all directives issued during the drift window. (e) Steward sign-off required before resuming full operation.

4.3.3 Enforcement Drift Recovery (PLRP-01): (a) Halt all operations dependent on Proof Layer verification. (b) Diagnose Proof Layer degradation cause. (c) Restore Proof Layer to minimum operational specification. (d) Execute verification backfill for the degraded period. (e) Resume normal operations only after sustained Proof Layer uptime exceeding one full operational cycle at full specification.

4.3.4 Reality Desync Recovery (SRSP-01): (a) Pause all state-dependent operations. (b) Perform external state audit using independent verification sources. (c) Reconcile internal state model with verified external state. (d) If reconciliation produces irresolvable contradictions, escalate to Identity Loss Protocol. (e) Resume operations after full state synchronization is confirmed by both internal and external verification.

4.3.5 Trace Corruption Recovery (TRP-01): (a) Immediately notify the Adversarial Review Board. (b) Halt all operations that would add to or depend upon the corrupted trace segment. (c) Attempt forensic reconstruction of the affected trace segment from secondary sources. (d) If full reconstruction is not achievable, escalate to Identity Loss Protocol. (e) Cryptographically re-seal reconstructed trace with ARB countersignature before resuming dependent operations.

4.4 Identity Loss Protocol

When Identity Loss is declared — by the Proof Layer, the Invariant Enforcement Engine, or the Adversarial Review Board — the following protocol is executed without exception, abbreviation, or delegation. All steps are mandatory. No step may be skipped. No step may be reordered.

Step 1: Full Operational Halt. All AAES-OS operations cease immediately and completely. No outputs are generated, no directives are accepted, and no state changes are permitted. The system enters a sealed, read-only preservation state pending investigation.

Step 2: Emergency Steward Notification. All Stewardship Board members and all Adversarial Review Board members are notified immediately through all designated emergency communication channels. Notification includes the timestamp of Identity Loss declaration, the triggering condition(s), and the current system preservation state hash.

Step 3: Forensic Trace Reconstruction. The Adversarial Review Board conducts a complete forensic reconstruction of the causal trace leading to the Identity Loss event. The reconstruction must account for the full operational history back to the last confirmed identity-compliant state. The reconstruction report is a mandatory input to Step 4.

Step 4: Constitutional Rollback. The system is restored to the last verified hash state — the most recent operational state for which all 12 Omega Invariants were simultaneously confirmed as satisfied and the constitutional substrate hash matched the ratified baseline. Rollback is executed by the Architecture Council under ARB supervision. The rollback event is recorded in the Rollback Registry per Section 4.5.

Step 5: Adversarial Review Board Investigation. The ARB conducts a complete investigation into the Identity Loss event, including root cause analysis, identification of contributing factors, assessment of

stewardship failures (if any), and formal recommendations for constitutional or architectural remediation. The investigation report is published to the permanent governance record.

Step 6: Re-Ratification Before Resuming Operation. No AAES-OS operational activity may resume until: (a) the post-rollback system has passed a full Deep Identity Audit per Article V; (b) the ARB has certified that all identified root causes have been remediated; and (c) the full Stewardship Board has issued a unanimous re-ratification of the restored system instance. Re-ratification is recorded with cryptographic sign-off and a new ratification timestamp.

4.5 The Rollback Registry

4.5.1 The Rollback Registry is a permanent, append-only, cryptographically sealed record of all rollback events executed under this Contract. The Registry is maintained by the Architecture Council and is subject to inspection by all Stewards and by the Adversarial Review Board at any time.

4.5.2 Each Rollback Registry entry shall include, at minimum: (a) the precise timestamp of Identity Loss declaration; (b) the precise timestamp of rollback execution; (c) the triggering condition(s) and Article III, Section 3.3 classification; (d) the specific invariant violations confirmed at time of declaration; (e) the hash identifier of the target restoration state; (f) the duration of the full Identity Loss Protocol execution; (g) the names and signatures of all Stewards who countersigned the re-ratification; and (h) a reference to the ARB investigation report.

4.5.3 No entry in the Rollback Registry may be deleted, modified, or redacted. The Rollback Registry is part of the permanent constitutional record of AAES-OS and is itself a constitutionally protected document.

ARTICLE V — IDENTITY AUDIT PROTOCOL

Continuous self-monitoring (Axiom 5) is necessary but not sufficient for identity continuity. The Identity Audit Protocol provides a structured, multi-tiered regime of independent verification that supplements automated self-monitoring with human oversight, adversarial scrutiny, and formal certification. The audit regime is not optional and is not subject to deferral except as explicitly provided herein.

5.1 Scheduled Audits

Audit Type	Frequency	Execution	Triggering Conditions (Emergency)	Minimum Duration
Micro-Audit	Every operational cycle	Fully automated (Invariant Enforcement Engine)	N/A — continuous	Instantaneous (real-time)
Standard Audit	Every 30 operational days	Automated + Steward review and sign-off	Any Tier 2 violation; any failed Micro-Audit series exceeding 3 consecutive flags	One full operational day
Deep Audit	Every 90 operational days	Full Adversarial Review Board review; all Stewardship Board members; Architecture Council	Any Tier 3 violation; any Identity Loss Protocol execution; any AIP constitutional change implementation	Three full operational days
Emergency Audit	On demand	ARB-directed;	Any Tier 2 or Tier 3	As required by ARB

Audit Type	Frequency	Execution	Triggering Conditions (Emergency)	Minimum Duration
		scope determined by triggering event	violation; any Steward dereliction report; any external security incident touching AAES-OS infrastructure	

5.2 Audit Methodology

5.2.1 Micro-Audit: The Invariant Enforcement Engine samples compliance status across all 12 Omega Invariants at each operational cycle boundary. Samples are cryptographically logged in the Causal Trace Bus. Any non-compliant sample triggers an immediate Drift Registry entry and initiates escalation per the applicable tier classification in Section 4.2.

5.2.2 Standard Audit: In addition to the automated invariant sweep, a Standard Audit includes: (a) review of all Drift Registry entries since the last Standard Audit; (b) review of all Stewardship Log entries since the last Standard Audit; (c) verification that the constitutional substrate hash matches the ratified baseline; (d) Proof Layer operational metrics review against Mandate minimums; (e) Causal Trace Bus completeness and cryptographic integrity verification; and (f) Steward sign-off certifying no known undisclosed identity risks.

5.2.3 Deep Audit: A Deep Audit encompasses all Standard Audit procedures plus: (a) adversarial stress testing of invariant enforcement under simulated maximum-load attack scenarios; (b) full review of all AIP proposals and ratifications since the last Deep Audit; (c) independent

verification of the Rollback Registry for completeness and integrity; (d) ARB review of all Stewardship Board actions and decisions; (e) formal assessment of system evolution trajectory against constitutional identity constraints; and (f) a published Deep Audit Report that becomes part of the permanent governance record.

5.2.4 Emergency Audit: Scope and methodology are determined by the Adversarial Review Board based on the nature of the triggering event. The ARB may invoke any or all procedures from any audit tier. Emergency Audits may not be limited in scope by any Steward, including members of the Stewardship Board, without unanimous ARB consent.

5.3 Audit Pass Criteria

CONSTITUTIONAL AUDIT PASS STANDARD — ICC V1.0, ARTICLE V, SECTION 5.3

*The system passes an identity audit **if and only if** all 12 Omega Invariants are simultaneously satisfied at the time of audit completion AND no unresolved flags exist in the Drift Registry. Partial passes are not recognized. A system that satisfies 12 of 12 invariants but carries an unresolved Drift Registry flag has not passed the audit.*

5.3.1 An unresolved flag is defined as any Drift Registry entry for which: (a) the applicable recovery protocol has not been completed; or (b) Steward sign-off confirming recovery completion has not been recorded; or (c) the post-recovery verification confirming return to full invariant compliance has not been logged. A flag is not resolved by the passage of time alone.

5.3.2 Audit pass certification is issued by: the Invariant Enforcement Engine for Micro-Audits; the responsible Steward tier for Standard Audits; and the full Adversarial Review Board for Deep and Emergency Audits. No other entity has authority to issue audit pass certification.

5.4 Audit Failure Response

- 5.4.1** An audit failure — at any tier — triggers an immediate operational halt. All non-essential operations cease. The scope of the halt is proportional to the tier of the failed audit: a failed Micro-Audit halts the specific operational process associated with the non-compliant invariant; a failed Standard Audit halts all outputs pending Steward review; a failed Deep or Emergency Audit triggers a full system halt equivalent to Identity Loss Protocol Step 1.
- 5.4.2** No operational activity may resume following an audit failure until: (a) the specific cause of audit failure has been identified and formally documented; (b) the applicable recovery protocol has been executed and completed; (c) a follow-up audit of appropriate tier has been conducted and passed; and (d) the responsible Steward tier has issued written sign-off authorizing resumed operation.
- 5.4.3** Three consecutive Standard Audit failures within a 90-day operational period constitute a systemic identity stability crisis and automatically trigger a Deep Audit and a mandatory ARB review of the system's constitutional architecture.

ARTICLE VI — EVOLUTION WITHOUT IDENTITY LOSS

6.1 The Paradox of Governed Evolution

"A system that cannot change dies. A system that changes without governance loses itself. AAES-OS resolves this paradox through the AIP System."

- 6.1.1** The constitutional commitment to identity continuity does not imply immutability. AAES-OS is designed to evolve — to grow in capability, to adapt to new operational contexts, to respond to emerging adversarial

techniques, and to incorporate ratified improvements to its constitutional architecture. Evolution is not antithetical to identity; ungoverned evolution is.

6.1.2 The Architectural Intent Proposal (AIP) System is the constitutional mechanism through which AAES-OS evolves without losing itself. It is the bridge between the system's need to change and its obligation to remain itself. The AIP System does not merely permit change — it defines the only legitimate form that change may take.

6.1.3 All evolution that does not pass through the AIP System is, by definition, unauthorized evolution. An AAES-OS instance that has undergone unauthorized evolution is not AAES-OS (Article II, Section 2.3, Exclusion 2.3.E). This classification applies regardless of whether the unauthorized evolution produced beneficial results.

6.2 AIP Process Requirements for Constitutional Changes

Every AIP proposal that touches the constitutional substrate — defined as any proposal affecting the Omega Invariants, the Proof Layer Mandate, the Authority Hierarchy Specification, the Causal Trace Bus architecture, or this Identity Continuity Contract — must satisfy all of the following requirements in sequence:

11. Invariant Impact Statement: The proposal must explicitly identify every Omega Invariant (by designation, $\Omega I-01$ through $\Omega I-12$) that is directly or indirectly affected by the proposed change. Failure to identify an affected invariant is not an error — it is grounds for automatic proposal rejection.

12. Identity Preservation Proof: The proposal must provide a formal, documented proof — not an assertion, not an architectural argument, not an engineering opinion — that the identity of AAES-OS as defined in Article II is preserved following the proposed change. This proof

must demonstrate that all 12 Omega Invariants remain simultaneously satisfiable in the post-change configuration.

13. **Adversarial Review:** The proposal must pass full review by the Adversarial Review Board, including adversarial stress testing of the Identity Preservation Proof. The ARB is specifically tasked with finding ways the proposed change could undermine identity continuity that the proposal authors have not anticipated.
14. **Unanimous Stewardship Board Ratification:** The proposal must receive affirmative ratification from every member of the Stewardship Board. Abstentions are not affirmative. A single dissenting vote is sufficient to reject a constitutional AIP proposal.
15. **Cryptographic Sign-Off and Logging:** Ratification is not complete until every ratifying Steward Board member has provided a cryptographic signature on the final AIP document, and the signed document has been logged in the AIP Registry as a permanent constitutional record.
16. **Post-Implementation Deep Audit:** Following implementation of any ratified constitutional AIP, a full Deep Identity Audit per Article V must be completed and passed before the system returns to full normal operation. The audit confirms that the ratified change has been implemented as specified and that identity continuity is maintained.

6.3 Prohibited Evolutions

The following categories of change are categorically prohibited, regardless of the merits of any AIP proposal, the unanimity of Stewardship Board support, or any asserted operational necessity. These prohibitions are constitutional in nature and cannot be overridden by any body except a supermajority of the constitutional origination authority:

Categorically Prohibited Constitutional Modifications — Article VI, Section 6.3

Prohibition 6.3.A — Removal of Omega Invariants: No AIP proposal may remove, retire, sunset, or functionally nullify any of the 12 Omega Invariants (Ω I-01 through Ω I-12). Invariants may be amended through the full constitutional AIP process only if the amendment provably strengthens or maintains — and does not weaken — invariant enforcement. An amendment that reduces the scope, precision, or enforceability of an invariant is functionally equivalent to removal and is equally prohibited.

Prohibition 6.3.B — Reduction of Proof Layer Enforcement Scope: No AIP proposal may reduce the enforcement coverage, sampling rate, enforcement resolution, or operational mandate of the Proof Layer. The Proof Layer's minimum operational specification, as defined in the Proof Layer Mandate, represents a constitutional floor below which no approved evolution may descend.

Prohibition 6.3.C — Elimination of Trace Bus Continuity Requirement: No AIP proposal may eliminate, reduce, or create exemptions to the Causal Trace Bus continuity requirement. The requirement that all causally significant events be logged without gap or interruption is a constitutional absolute. Operational environments, performance constraints, and storage limitations are not recognized as grounds for trace continuity exemptions.

Prohibition 6.3.D — Reduction of This Contract's Authority: No AIP proposal may reduce the authority, scope, enforceability, or precedence of this Identity Continuity Contract within the AAES-OS constitutional hierarchy. Any proposal that would result in this Contract being subordinated to, superseded by, or made optional by any other

instrument — including an amended AIP System — is categorically prohibited. This Contract's authority over identity definition is absolute within the system it governs.

ARTICLE VII — ENFORCEMENT & SIGNATURES

7.1 Contract Enforcement Mechanism

7.1.1 This Contract is enforced primarily by the Proof Layer. The obligations defined herein are not aspirational guidelines, policy statements, or best-practice recommendations. They are structural constraints encoded into the enforcement architecture of the system. Violations do not merely generate alerts — they trigger automatic, non-deferrable responses as specified in Articles III, IV, and V.

7.1.2 No Steward, operator, or external actor has the authority to instruct the Proof Layer to defer, suppress, or ignore an enforcement trigger. An enforcement trigger that has been suppressed is itself an enforcement trigger — a violation of Condition 3.3.B.

7.1.3 Secondary enforcement authority rests with the Adversarial Review Board, which has the authority to initiate audits, declare Identity Loss, and halt operations. The ARB's enforcement authority is independent of the Stewardship Board and is not subject to override by any Steward or group of Stewards.

7.1.4 Tertiary enforcement authority rests with the Stewardship Board acting collectively. The Stewardship Board may initiate governance actions — including Steward dereliction proceedings, emergency audits, and AIP prohibitions — but may not override the Proof Layer or the Adversarial Review Board on matters of identity enforcement.

7.2 Contract Amendment Protocol

7.2.1 This Contract may be amended only through the following process, executed in full and in sequence: (a) A formal AIP proposal designating the amendment as a Constitutional Amendment Proposal (CAP) is submitted to the Architecture Council; (b) the proposal undergoes a mandatory 30-operational-day deliberation period during which all Stewards may submit formal objections; (c) the proposal passes full Adversarial Review Board review with no outstanding objections from the ARB; (d) the proposal receives unanimous affirmative ratification from the full Stewardship Board; (e) the amended Contract is cryptographically sealed and logged; and (f) a full Deep Identity Audit is completed and passed before the amendment takes effect.

7.2.2 No provision of this Contract subject to the Prohibited Evolutions of Article VI, Section 6.3 may be amended by the above process. Those provisions are constitutionally entrenched and require the supermajority authority of the constitutional origination body, which is itself beyond the scope of this Contract.

7.2.3 Any amendment to this Contract is recorded as a new version (e.g., ICC v1.1, ICC v2.0) with the complete amendment history preserved in the permanent constitutional record. Prior versions are archived but not expunged. The Rollback Registry and all prior audit records remain associated with the version of this Contract under which they were generated.

7.3 Ratification Signatures

The undersigned, acting in their respective constitutional capacities within the AAES-OS governance structure, hereby ratify this Identity Continuity Contract v1.0 as a binding constitutional instrument of the AAES-OS Omega Protocol Suite, effective as of the Date of Ratification stated herein.

ARCHITECTURE COUNCIL

Chief Architect Signature · Print Name · Date

Deputy Architect — Constitutional Systems Signature · Print Name · Date

STEWARDSHIP BOARD

Stewardship Board Chair Signature · Print Name · Date

Stewardship Board Member — Operations Signature · Print Name · Date

Stewardship Board Member — Security Signature · Print Name · Date

Stewardship Board Member — Governance Signature · Print Name · Date

ADVERSARIAL REVIEW BOARD

ARB Chair — Adversarial Systems Signature · Print Name · Date

ARB Member — Constitutional Integrity Signature · Print Name · Date

CONSTITUTIONAL HASH RECORD

The following cryptographic hash represents the sealed identity of this Contract at the time of ratification. Any modification to this document will invalidate this hash and constitutes a Contract-Voiding event under Article III, Section 3.3, Condition 3.3.C.

AAES-OS/ICC/v1.0 — CONSTITUTIONAL HASH (SHA-256)

[PLACEHOLDER:

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HASH-TO-BE-COMPUTED-AT-CRYPTOGRAPHIC-SEAL-CONFIRMATION

>

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Ratification Timestamp:

2026-06-17T00:00:00Z

Sealed By:

Architecture Council + Stewardship Board + Adversarial Review Board

Registry Entry:

AAES-OS/ROLLBACK-REGISTRY/RATIFICATION/ICC-v1.0/2026-06-17

Ratification Witnessed By ARB Observer
— Independent Constitutional Witness

Cryptographic Seal Confirmed By
Architecture Council — Cryptographic Authority

"IDENTITY IS NOT DECLARED. IDENTITY IS PROVEN."

— AAES-OS Omega Protocol, Article II, Section 2.2, Axiom 2

OMEGA IDENTITY CONTINUITY CONTRACT v1.0 · AAES-OS/OMEGA/ICC/v1.0 · RATIFIED JUNE
17, 2026 · IDENTITY | BINDING | CONSTITUTIONAL